

Public Key Cryptography: RSA

RSA is an asymmetric encryption algorithm. This means it uses two different keys:

1. **Public Key:** Can be shared with anyone. Used to encrypt data.
2. **Private Key:** Must be kept secret. Used to decrypt data.

The Math: RSA security relies on the difficulty of factoring very large prime numbers. If you multiply two large primes together, it is easy to get a product, but extremely hard to work backward to find the original primes.

Digital Signatures: RSA can also be used for signatures. If a sender encrypts a hash of a message with their *private* key, anyone with the *public* key can verify that the message truly came from them.